

AI531C — Week 9 Pre-Class Activity (PCM-9: How Far Can One Neuron See?)

Prepares: Lecture 23 (Receptive fields, Tue 6 Oct) · Lectures 24–25 (Feature maps; Data pipeline & augmentation), Wed 7 Oct · Tutorial T9 (Tue 6 Oct).

1 · Topic

The receptive field: how stacking small kernels lets a deep unit see a wide patch — and why depth, stride and dilation are three different ways to buy reach.

2 · Why you are preparing this

Lecture 23 turns RF into arithmetic you can run. Arriving with the recurrence already applied once means class can debate DESIGN: how much field does a task need, and what is the cheapest way to get it?

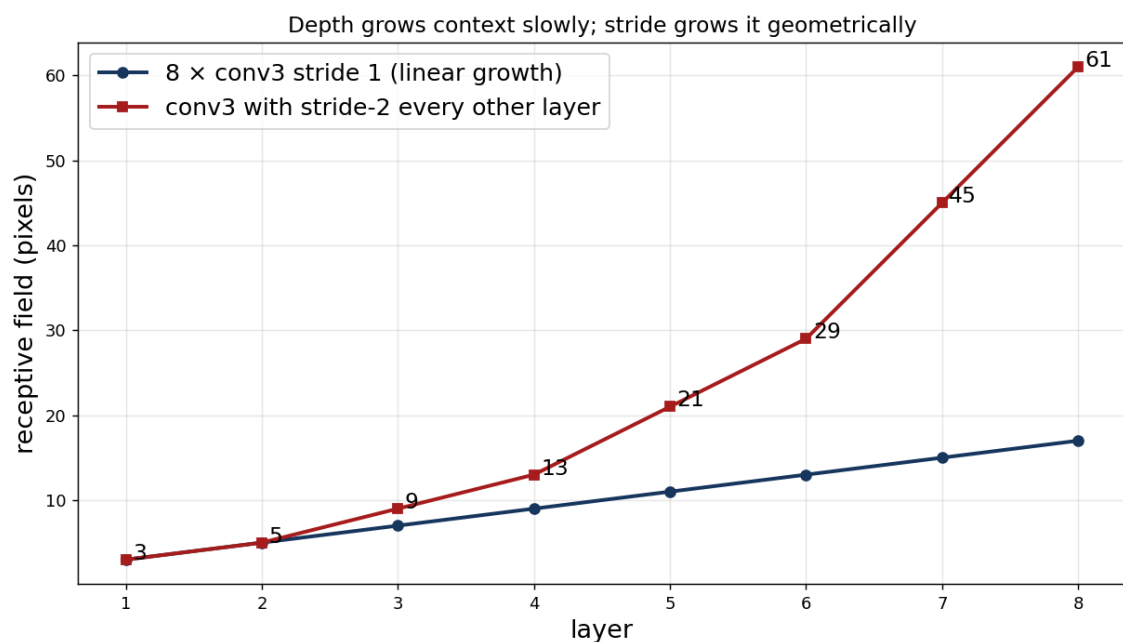
3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 23 chapter, the sections on the RF recurrence and the growth table (about 3 pages);
- Lecture 25 chapter, ONLY the image-level split rule (one page — the leakage trap).

Optional depth: Prince, receptive-field figures (skim).

4 · Interpret one figure (4 min)



Source: course figure, Lecture 23 (receptive-field growth).

In two sentences: read from the plot how fast the RF grows for plain 3×3 stacking vs with stride-2 layers inserted — which curve bends, and why?

5 · One small calculation (5 min)

Two stacked 3×3 convs (stride 1): compute the receptive field. Add a third. General rule for k layers?

6 · Predict (4 min)

Predict: to judge whether a 200-px-wide object is fully 'seen' by one output unit, roughly how many plain 3×3 layers would you need — and is that practical without stride or dilation?

7 · Inspect this fragment (3 min)

```
rf = 1
for k, s in layers: # (kernel, stride)
    rf = rf + (k - 1) * jump; jump *= s
# what is 'jump' tracking?
```

Question: in the recurrence, what does the jump variable represent physically on the input grid?

8 · Bring one question to class

One question about receptive fields — or your capstone's largest object size in pixels (bring the number; T9 sizes an encoder for it).

Submit: items 4-6 — half a page maximum, before Tue 6 Oct, 9:00 AM.

Expected time: 30-35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2...PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.